



DSA MARATHON

Day 2 (Tuesday) — Practice Questions

Level: Easy → Moderate (Concept Building)

Part A — 15 Conceptual MCQs with Answers

Q1. What happens if you write `cout << 5/2; ?`

- A) 2.5 B) 2 C) 3 D) Compile Error

Answer: B

Q2. Which statement correctly reads a full name after reading an integer?

- A) `getline()` directly B) `cin.ignore(); getline(...)` C) `fflush(stdin)` D) `scanf()`

Answer: B

Q3. Which declaration is valid?

- A) `int 2x;` B) `int _count;` C) `int float;` D) `int class;`

Answer: B

Q4. Which expression changes `x` from 8 to 13?

- A) `x=x+5` B) `x+=5` C) Both A and B D) `x++`

Answer: C

Q5. Which expression prints 6 when `x=5`?

- A) `x++` B) `++x` C) `x--` D) `--x`

Answer: B

Q6. Which is true about `x++`?

- A) Increment then use B) Use then increment C) Decrease D) Nothing

Answer: B

Q7. Which operator has higher precedence?

A) && B) == C) + D) *

Answer: D

Q8. Output of: `int a=10; cout<<(a>5 && a<20);`

A) 0 B) 1 C) 10 D) Error

Answer: B

Q9. ASCII difference between 'a' and 'A' is

A) 26 B) 31 C) 32 D) 64

Answer: C

Q10. Character '9'-'0' gives

A) 9 B) 57 C) 48 D) 1

Answer: A

Q11. `static_cast<double>(5)/2` produces

A) 2 B) 2.0 C) 2.5 D) 3

Answer: C

Q12. Which operator cannot be used with double values?

A) + B) % C) / D) *

Answer: B

Q13. Which line is a comment?

A) `#comment` B) `//comment` C) `/comment/` D) `**comment**`

Answer: B

Q14. `sizeof(char)` is typically

A) 1 byte B) 2 bytes C) 4 bytes D) 8 bytes

Answer: A

Q15. Value of $10+6/2*3-4$

A) 12 B) 15 C) 19 D) 24

Answer: B

Part B — 10 Coding Questions (Easy → Moderate) with Answers

Q1. Read three integers and print their average as a decimal.

```
int a,b,c;
cin>>a>>b>>c;
double avg=static_cast<double>(a+b+c)/3;
cout<<avg;
```

Q2. Swap two numbers without using a third variable (using arithmetic).

```
int a,b;
cin>>a>>b;
a=a+b;
b=a-b;
a=a-b;
cout<<a<<" "<<b;
```

Q3. Read a character and print whether it is uppercase or lowercase using ASCII.

```
char ch;
cin>>ch;
if(ch>='A' && ch<='Z') cout<<"Uppercase";
else if(ch>='a' && ch<='z') cout<<"Lowercase";
```

Q4. Convert a lowercase letter to uppercase using ASCII.

```
char ch;
cin>>ch;
cout<<char(ch-32);
```

Q5. Convert an uppercase letter to lowercase.

```
char ch;
cin>>ch;
cout<<char(ch+32);
```

Q6. Read an integer and print its square and cube.

```
int n; cin>>n;
cout<<n*n<<" "<<n*n*n;
```

Q7. Demonstrate difference between ++x and x++.

```
int x=5;
cout<<++x<<" ";
cout<<x++;
```

Q8. Print quotient and remainder of two numbers.

```
int a,b;  
cin>>a>>b;  
cout<<a/b<<" "<<a%b;
```

Q9. Read first name and last name separately and print Full Name.

```
string f,l;  
cin>>f>>l;  
cout<<f<<" "<<l;
```

Q10. Evaluate expression using precedence: $(8+4)/2*3+5$

```
cout<<(8+4)/2*3+5; //23
```

Thinking Questions

Reflect on these — no single fixed answer, but strong reasoning is the goal.

1. Why is `static_cast` preferred over C-style casting?
2. Explain a real situation where `getline()` is better than `cin`.
3. Why should parentheses be used even when you know operator precedence?

End of Day 2 Practice Set

NextGen Data Minds